

BIT 4543 — ARTIFICIAL INTELLIGENCE GROUP PROJECT

AI Sentiment Analysis Dashboard

Classifying Amazon customer reviews with TF-IDF and Logistic Regression

Positive

Neutral

Negative

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Background & Problem Statement

Online shopping has led to a huge number of customer reviews being published on the Web. However, it becomes impractical to read each and every one of them to understand the overall sentiment and opinion.

Reviews also come in messy with informal language and different types of errors like wrong spellings, HTML code, and sentiment words that change directions in the same sentence.

Besides, the major dilemma is in selecting a method that not only performs well but does it without the burden in terms of the deep learning or the transformer model cost and complexity.

So: can a lightweight TF-IDF + Logistic Regression pipeline classify Amazon reviews well enough to power a real, interactive dashboard?

SCALE OF THE CHALLENGE

2

Product categories analysed

100K

Reviews sampled (50K per category)

3

Sentiment classes: negative, neutral, positive

Aim & Objectives

Aim: Develop and evaluate a machine-learning system that automatically classifies Amazon reviews as positive, neutral, or negative.

1

Build an AI app that classifies review text into 3 sentiment classes

2

Collect & clean a large sample from Video Games and All Beauty categories

3

Train category-specific TF-IDF + Logistic Regression models

4

Visualise sentiment distributions, ratings, and model performance

5

Create a real-world interface while allowing users to optionally provide feedback.

6

Evaluate with classification metrics and user acceptance testing

Project Scope

Sentiment labels are derived directly from star ratings:

1-2 ★

Negative

3 ★

Neutral

4-5 ★

Positive

In scope

- ▶ Category selection & review prediction
- ▶ Probability display for each class
- ▶ Dataset charts & model metrics
- ▶ Analysis history & feedback recording

Out of scope

- ▶ User login & real time Amazon data collection
- ▶ Aspect based sentiment explanation
- ▶ Multilingual modelling
- ▶ Automated business decision making, file upload

Literature Review Summary

Study	Domain	Approach	Relevance
Adak et al. (2022)	Food-delivery reviews	ML, DL & XAI review	Background on approaches & explainability
Chinnalagu & Durairaj (2021)	Customer reviews	Linear ML, FastText, neural	Supports linear models for text classification
Hashmi & Yayilgan (2024)	Amazon product reviews	TF-IDF, Word2Vec, FastText, ML/DL/transformers	Directly relevant Amazon review sentiment
Shaik Vadla et al. (2024)	Amazon reviews	BERT/T5, aspect-based	Demonstrates transformer approaches

Research gap: Advanced deep-learning and transformer approaches can boost accuracy, but need heavier compute and implementation effort. This project targets a practical, interpretable middle ground TF-IDF + Logistic Regression evaluated experimentally against simpler baselines.

End-to-End Methodology



Same workflow applied independently to both Video Games and All Beauty categories, so each gets its own tuned model.

80/20

stratified train/test split

30,000

TF-IDF features (uni + bigrams)

**Logistic
Regression**

final selected classifier

Dataset & Preprocessing Yield

Video Games

48,371

final reviews used for modelling
(from 50,000 sampled)

All Beauty

48,443

final reviews used for modelling
(from 50,000 sampled)

Video Games cleaning pipeline

Sampled: 50,000

– missing text: 49,997

– duplicates: 48,476

– empty after cleaning: 48,375

WHY PRESERVE NEGATION?

"not good" should never be reduced to just **"good"**.

Preserved words: not, no, never, nor, neither, hardly, barely, nothing
kept through stop word removal and lemmatization.

Text Preprocessing Pipeline

1

Remove missing / duplicate reviews

2

Lowercase all text

3

Strip HTML tags & URLs

4

Remove punctuation & numbers

5

Collapse excess whitespace

6

Remove English stop words

7

Preserve negation terms

8

WordNet lemmatization

9

Drop reviews empty after cleaning

TF-IDF Feature Extraction

Parameter	Value
Maximum features	30,000
N-gram range	1–2 (unigrams + bigrams)
Min document frequency	2
Max document frequency	0.95
Sublinear TF	Enabled

N-GRAMS CAPTURE MORE MEANING

"good"



"very good"



"not good"



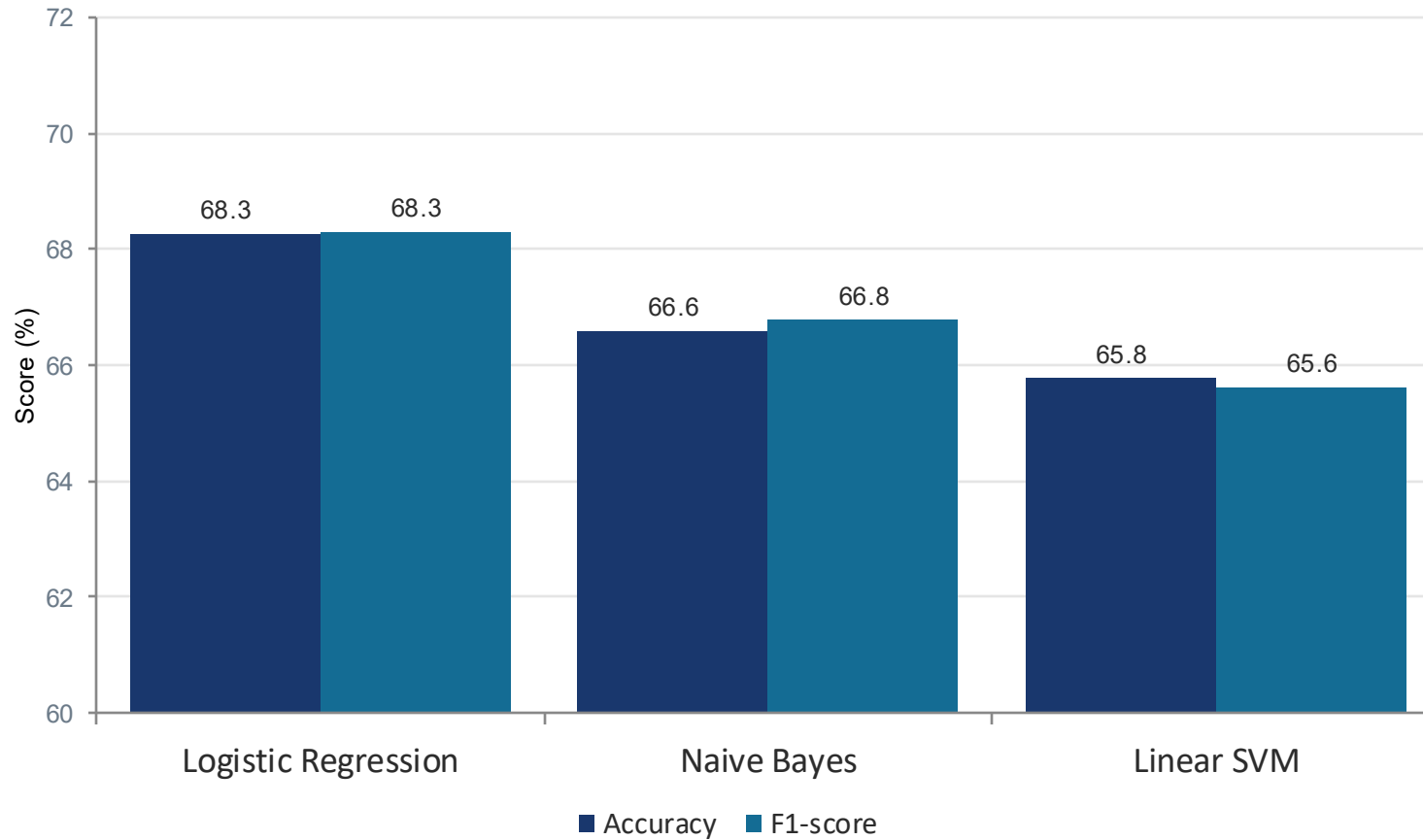
"great game"



TF-IDF was fit only on training data, then applied to the test set preventing test data leakage into the vocabulary.

Comparing Classification Models

Video Games dataset Accuracy vs. F1-score



WHY LOGISTIC REGRESSION

- Highest accuracy & F1 across both categories
- Handles high-dimensional sparse TF-IDF features efficiently
- Outputs class probabilities, not just a label
- Simple, fast to train, easy to deploy

Final Model Performance

Video Games

68.34%

Accuracy

68.37%

Weighted F1

All Beauty

69.33%

Accuracy

69.28%

Weighted F1

Video Games perclass breakdown

Sentiment	Precision	Recall	F1-score
Negative	0.70	0.72	0.71
Neutral	0.58	0.59	0.58
Positive	0.77	0.74	0.76

Neutral is hardest to classify it overlaps with weak positive and negative expressions, pulling down its precision and recall.

Confusion Matrix Video Games

Actual (rows) vs. Predicted (columns)

	Negative	Neutral	Positive
Negative	2,389	760	150
Neutral	822	1,913	529
Positive	200	609	2,303

Reading it: diagonal cells (bold) are correct predictions. Most confusion happens between Neutral and its neighbours 822 neutral reviews were predicted negative and 529 were predicted positive.

The Streamlit Dashboard

Core features

Select product category

Enter a review and get an instant sentiment prediction

View class probabilities

Explore processed dataset charts & model metrics

Browse analysis history and submit feedback

EXAMPLE PREDICTION

"Absolutely love this best purchase all year!"



Deployed on Streamlit Community Cloud, connected to GitHub updates to the main branch trigger redeployment automatically.

User Acceptance Testing

Tester	Tasks Passed	Tasks Failed	Status
Tester 1	8/9	1	Fail/Partial
Tester 2	8/9	1	Fail/Partial
Tester 3	9/9	0	Pass

Tasks covered: open app, select category, submit positive/negative/blank reviews, switch category, open dashboard & history, submit feedback.

KEY OBSERVATIONS FROM TESTING

- Long, mixed reviews sometimes misclassified as neutral
- Sarcasm not reliably understood read as positive
- Slang / abbreviations occasionally missed
- Core flow predict, switch category, dashboard, history, feedback worked smoothly for all testers

Limitations & Future Improvements

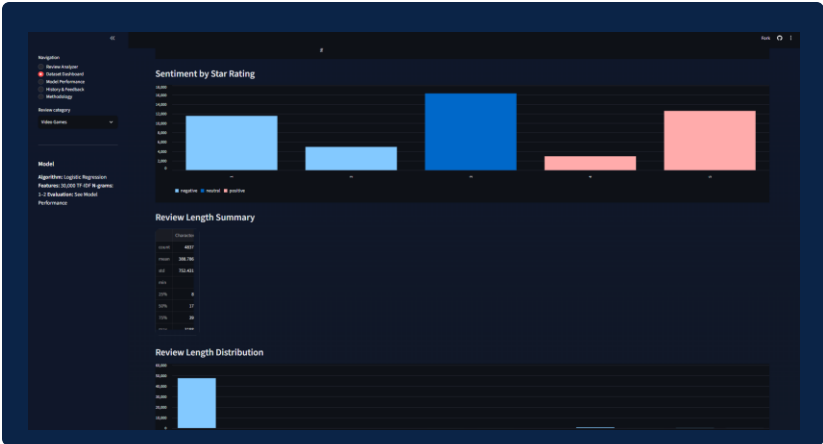
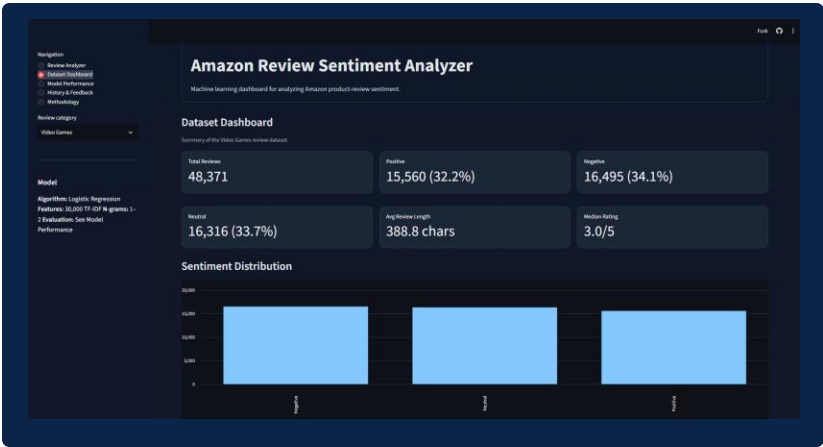
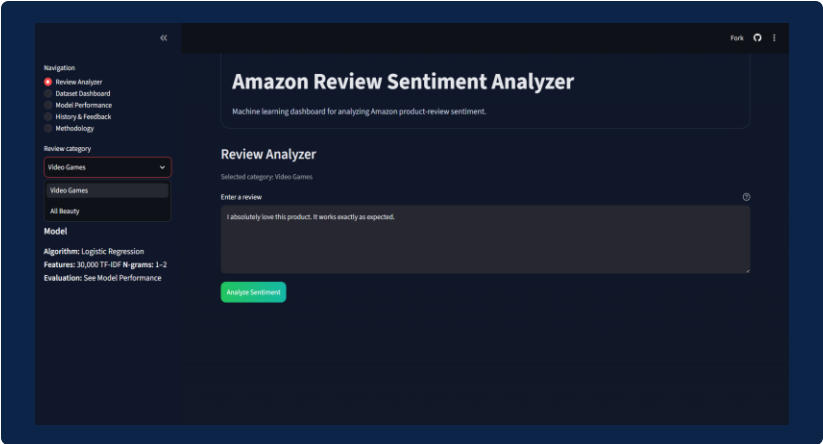
LIMITATIONS

- Labels inferred from star ratings, not manual annotation
- English language reviews only
- TF-IDF has limited contextual understanding struggles with sarcasm & mixed sentiment
- Results based on a single 80/20 split
- Model may not generalise well to unseen product categories

FUTURE IMPROVEMENTS

- Transformer models BERT, RoBERTa for richer context
- Aspect-based sentiment analysis graphics, price
- Explainability via LIME or SHAP
- Larger, more diverse datasets and categories
- Cross-validation and deeper error analysis

SCREENSHOT OF APP



SCREENSHOT OF APP



Navigation

- Review Analyzer
- Dataset Dashboard
- Model Performance
- History & Feedback
- Methodology

Review category

- Video Games

Model

- Algorithm: Logistic Regression
- Features: 30,000 TF-IDF bigrams 1-2
- Evaluation: See Model Performance

Amazon Review Sentiment Analyzer

Machine learning dashboard for analyzing Amazon product review sentiment.

Model Performance

Evaluation of the Video Games 30K TF-IDF 1-Logistic Regression model

Accuracy68.34%

Precision68.44%

Recall68.34%

F1 Score68.37%

Classification Report

	precision	recall	F1 score	support
negative	0.7088	0.7088	0.7088	3399
neutral	0.5045	0.5052	0.5048	3394
positive	0.7719	0.7684	0.7698	3312

Confusion Matrix

	Predicted Negative	Predicted Neutral	Predicted Positive
Actual Negative	2396	765	238
Actual Neutral	825	2355	515
Actual Positive	395	813	2044

The confusion matrix shows how many reviews were correctly or incorrectly classified for each sentiment class.

Classifier Comparison

Ref:python -m evaluate_metrics.py ModelA to generate a reproducible baseline, Naive Bayes, and Logistic Regression comparisons.

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Machine learning dashboard for analyzing Amazon product review sentiment.

History & Feedback

Submitted reviews and optional feedback are stored in this app's local SQLite database. We can enter confidential or personal information.

Analysis ID	Created (UTC)	Category	Review	Prediction	Confidence	Feedback
9	2024-09-17 11:34:23+00:00	Video Games	I absolutely love this product, it works exactly as expected.	Positive	94.00%	No feedback
8	2024-09-17 10:09:49+00:00	Video Games	disent games, bought on xbox	Neutral	89.87%	Correct
7	2024-09-17 10:09:23+00:00	Video Games	I hate this game, refunded	Negative	75.93%	No feedback
6	2024-09-17 10:09:09+00:00	Video Games	I absolutely love this product, it works exactly as expected.	Positive	94.00%	No feedback

Add or update feedback

Analysis to review

#9: 2024-09-17 11:34:23+00:00 - Video Games - Positive

Was the predicted sentiment correct?

No

Yes

Corrected sentiment if the prediction was incorrect

Not applicable

Optional feedback comment

Save feedback

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Machine learning dashboard for analyzing Amazon product review sentiment.

Methodology and limitations

Labels: Star ratings are used as a sentiment proxy: 1-3 stars are negative, 4 stars are neutral, and 4-5 stars are positive.

Preprocessing: The workflow involves missing and duplicate reviews, stops HTML and URLs, lowercases text, removes stop words while preserving key negation terms, and tokenizes the remaining tokens.

Evaluation: Data is split into shuffled 80/20 train and test sets with a fixed random seed. TF-IDF forms its vocabulary from training text only, preserving test set feature leakage.

Local database: Submitted review text, category, prediction, confidence, and optional feedback are stored in SQLite for the coursework demonstration.

Limitations: Rating-derived labels can be noisy for mixed or sarcastic reviews. TF-IDF models also have limited contextual understanding and may not generalize to other languages or product categories.

CONCLUSION

A Practical Approach That Works

TF-IDF + Logistic Regression delivers a reproducible, interpretable, and deployable sentiment-analysis pipeline without the cost of deep learning or transformer models.

68.34%

Video Games accuracy

69.33%

All Beauty accuracy

68.37%

Video Games F1-score

3/3

Testers rated the app usable

Traditional machine learning, applied thoughtfully, still delivers real practical value.

Thank you